Richard Lim

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Summary

Final-year undergraduate at Bowdoin College majoring in Mathematics and Computer Science. I study the geometry of generalization in deep learning with a focus on length extrapolation for combinatorial optimization targets using tropical, polyhedral, and optimization geometry.

Education

May 2027 **Bowdoin College**, Brunswick, ME GPA: 4.0/4.0
Bachelor of Arts, Computer Science and Mathematics
Relevant Coursework: Intro Analysis; Topology; Computational Complexity; Multivariable Calculus; Linear Algebra; Artificial Intelligence; Combinatorics and Graph Theory; Advanced Analysis (incl. measure theory and stochastic calculus, upcoming Spring 2027); Advanced Algebra (upcoming Spring 2027)

Spring 2026 **Eötvös Loránd University (ELTE)**, Budapest, HU With Honors (Kitüntetéssel)
Budapest Semesters in Mathematics
Relevant Coursework: Functional Analysis; Discrete and Convex Geometry; Quantum Logic and Probability (incl. spectral theory and tensor products)

Research Experience

Tropical (Min-Plus) Geometry of Length Generalization June 2026 - August 2026
Visiting Researcher, Naval Postgraduate School, Monterey, CA
Advisor(s): Dr. Ruriko Yoshida

- Studied why models trained on small dynamic programming problems may fail to generalize to larger instances, using tools from graph theory and convex geometry.
- Proved a hierarchy of ways larger problems can be assembled from smaller ones, showing that tropical-native series–parallel constructions capture only a restricted class of dynamic programs.
- Developed methods to determine when structurally different dynamic programs compute the same function and to reduce them to a canonical representation.
- Built BrainSpace, a PCA-based tool for visualizing OOD generalization in hidden state space.

Clustered Mini-batching and Sampling for SGD August 2025 - May 2026
Visiting Researcher, HUN-REN SZTAKI AI Lab, Budapest, HU
Advanced Independent Study, Bowdoin College
Advisor(s): Dr. Jeová Farias Sales Rocha Neto, Dr. Bálint Zoltán Daróczy, Dr. András Benczúr

- Designed and implemented k-Neighborhood Aware Sampling (k-NAS), a minibatch-construction algorithm that forms spatially-local batches by sampling around an anchor under a temperature-controlled softmax over pairwise distances, interpolating continuously between local and uniform sampling
- Ran a controlled study over three replacement-sampling schemes and varied temperature on a synthetic regression task, isolating the effect of geometric batch locality on optimization and generalization.

Sign2vec: An Expressive Embedding Space for ASL May 2025 - October 2025
Bowdoin Research Fellow, Bowdoin College
Advisor(s): Dr. Jeová Farias Sales Rocha Neto

– Adapted Spatio-Temporal Graph Convolutional Networks (ST-GCNs) to representation learning for American Sign Language (ASL), bridging computer vision and NLP.
– Trained and evaluated ML models on >30,000 videos using PyTorch and High Performance Computing, conducting experiments to optimize model performance.

Group Differences in Lexical Search and Retrieval May 2024 - November 2024
Gibbons Research Fellow, Lexicon Lab, Bowdoin College
Advisor(s): Dr. Abhilasha A. Kumar

– Created a Python-based computational model to study lexical retrieval using data science packages like NumPy and pandas, and NLP tools like the Universal Sentence Encoder (USE).
– Analyzed data from the Semantic Fluency Task (SFT) for >500 participants using the *forager* model, investigating lexical clustering and integrating vector embeddings with behavioral data.
– Wrote E-Prime program for use in electroencephalographic (EEG) studies on lexical retrieval.

Publications

Lim, **Richard F. M.** and Ruriko Yoshida (2026). *On the Representational Geometry of Dynamic Programs*. arXiv: 2608.25034 [cs.LG]. URL: https://arxiv.org/abs/2608.25034. Accepted at NeurReps 2026, Extended Abstracts Track.

Talks and Presentations

Invited Talks

Lim, **Richard F. M.** (July 2026). "Learning Algorithms, Not Answers: A Framework for Length-OOD Generalization on Dynamic Programming Problems". Workshop in Operations Research/Systems Analysis. Guest lecture. Monterey, CA. Invited by Dr. Ruriko Yoshida.

Lim, **Richard F. M.** (May 2026). "k-Neighborhood Aware Sampling (k-NAS) for manifold-aware learning". HUN-REN AI Symposium. Jointly organized with Nanyang Technological University, Singapore. Budapest, HU. Invited by Dr. András Benczúr.

Lim, **Richard F. M.** (Mar. 2026). "Clustered Mini-batching and Sampling for Stochastic Neural Network Optimization". Seminar on the Manifolds in Machine Learning. Budapest, HU. Invited by Dr. András Benczúr and Dr. Bálint Zoltán Daróczy.

Poster Presentations

Lim, **Richard F. M.** and Ruriko Yoshida (Aug. 2026). "Teaching Algorithms to Machines". Summer Internship Poster Session. Monterey, CA.

Lim, **Richard F. M.** and Jeová Farias Sales Rocha Neto (Oct. 2025). "Toward an Expressive Semantic Map for Sign Language Processing". President's Summer Research Symposium. Brunswick, ME.

Lim, **Richard F. M.**, Amanda Cooney, Channing Hambric, and Abhilasha A. Kumar (Nov. 2024). "Group Differences in Lexical Source Use and Clustering During Memory Search". Psychonomic Society 65th Annual Meeting. New York, NY.

Lim, **Richard F. M.** and Abhilasha A. Kumar (Oct. 2024). "CompCog: Modeling Search Within the Mental Lexicon". President's Summer Research Symposium. Brunswick, ME.

Grants and Funding

CXD Funded Internship Grant $7000, June 2026

Awarded for the study of Tropical methods for machine learning with Dr. Ruriko Yoshida at the Naval Postgraduate School in Monterey, CA

Bowdoin Research Award $5520, May 2025

Awarded for the study of NLP techniques for Sign Language understanding with Dr. Jeova Farias

Gibbons Summer Research Fellowship $5520, May 2024

Awarded for the study of lexical retrieval using computational models with Dr. Abhilasha Kumar

Academic Honors

Book Award 2024, 2025

Awarded for receiving a grade point average of 4.0 during the last academic year

Sarah and James Bowdoin Scholar (Dean's List) 2024, 2025

Awarded to the top 20% of all eligible students with the highest grade point average

Teaching Experience

Teaching Assistantships

Departments of Mathematics and Computer Science, Bowdoin College

Algorithms	Fall 2025, Fall 2026
Intermediate Integral Calculus	Fall 2024
Multivariable Calculus	Fall 2025
Linear Algebra	Spring 2025
Intermediate Linear Algebra	Fall 2026
Quantum Computing	Fall 2026

Quantitative Tutor Fall 2024 - Present

Baldwin Center for Learning and Teaching, Bowdoin College

Tutored students in a range of Mathematics and Computer Science classes

Leadership and Service

Reviewer 2026

NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations

Student Advisor for the Tenure-Track Faculty Search 2024 - 2025

Department of Computer Science, Bowdoin College

Captain, Bowdoin Quiz Bowl Fall 2025, Fall 2026 - Present

Managed club budget and led a team of 15 members in weekly practice and intercollegiate tournaments

Resident Assistant Fall 2024 - Spring 2025

Office of Residential Life, Bowdoin College

– Led orientation and community-building activities for 120 first-year and upper class residents

– Resolved conflicts and provided social and emotional support to a floor community of 16

Ammunition Technician January 2021 - November 2022
Singapore Armed Forces

– Planned duty schedules and food rations for the ammunition depot staff of 80
– Developed and executed orientation, and safety and security training programs for posted-in soldiers
– Awarded Best Soldier of the Month and Letter of Commendation from the Battalion CO